# Sai Siddharth Cilamkoti

+1-716-808-4577 | siddharthcilamkoti@gmail.com | linkedin.com/in/sai-siddharth-cilamkoti | github.com/siddharthc30 | siddharthc30.github.io

## Education

**University at Buffalo** — Buffalo, New York
*Master of Science in Computer Science* — *Aug. 2023 – May 2025*
- Concentration: Machine Learning and Artificial Intelligence
- Relevant Courses: Data Structures & Algorithms, Operating Systems, Machine Learning, Deep Learning

**Amrita Vishwa Vidyapeetham** — India
*Bachelor of Technology in Computer Science with AI* — *Aug. 2019 – May 2023*

## Experience

**Software Engineer (Capstone)** — Aug. 2024 – Dec. 2024
*Kaleida Health* — *Buffalo, New York*
- Architected a scalable Microservices Architecture for a full-stack healthcare platform using Java Spring Boot and React, deploying to Microsoft Azure infrastructure to ensure high availability for hospital operations.
- Designed and secured RESTful APIs using JWT-based authentication, role-based access control (RBAC), and HTTPS best practices to protect sensitive healthcare data.
- Engineered secure RESTful APIs and integrated WebSockets to support real-time, low-latency communication, successfully handling high volumes of concurrent connections.
- Optimized SQL database schemas and complex queries within MySQL, resulting in improved data retrieval speeds and reduced system latency.
- Led the backend development lifecycle (SDLC), implementing Object-Oriented Design principles to support multi-language translation features, driving a 25% increase in user engagement.

**Research Software Engineer** — Feb. 2024 – Present
*University at Buffalo* — *Buffalo, New York*
- Engineered a Distributed System evaluation pipeline for large-scale code generation, automating telemetry tracking and execution across 1,000+ distinct test cases.
- Implemented parallel execution strategies using Python to optimize runtime efficiency and monitor server-side resource utilization, identifying optimal performance-to-cost configurations.
- Devised backend optimization strategies (Quantization/Pruning) for high-load systems, achieving a 20% reduction in server energy consumption while maintaining system integrity.
- Contributed technical documentation and system analysis accepted for publication at FLLM 2025, focusing on infrastructure performance metrics.

**Machine Learning Engineer Intern** — May 2022 – Aug. 2022
*INAI, IIIT-Hyderabad* — *Hyderabad, India*
- Built and deployed a scalable REST API using Flask, enabling seamless integration of computer vision models into live production environments.
- Optimized backend inference pipelines to process real-time video data, improving overall system throughput and preprocessing speed by 8%.
- Enhanced simulation environment performance by fine-tuning underlying data processing logic in Python.

## Projects

**Plant Disease Detection with Chatbot Support** | *Python, PyTorch, LangChain, LlamaIndex* — GitHub | 2024
- Developed an AI system utilizing advanced computer vision techniques and fine tuned a VGG model to rapidly identify early-stage plant diseases through detailed leaf image analysis achieving a 92% accuracy.
- Integrated OpenAI's DaVinci LLM via LangChain to generate context-aware plant management advice for effective plant management and disease prevention strategies.

**Review Analyzer** | *Python, Flask, Azure Cognitive Services, REST API* — GitHub | 2023
- Engineered an ML-driven decision-support tool to analyze reviews and extract sentiment patterns for Amazon purchasing choices.
- Built a full-stack Flask application integrating Azure Cognitive Services, structured JSON APIs, and secure cloud-based inference endpoints.

## Technical Skills

**Programming**: Python, Java, SQL, Javascript, C++, HTML/CSS
**Development Environments and Tools**: Docker, Microsoft Azure, Git, Linux, Cursor
**Libraries & Frameworks**: PyTorch, TensorFlow, Sklearn, Numpy, Pandas, Selenium, Django, FastAPI, PostgreSQL, Spring, SpringBoot, React